

ERICK ROSS

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EDUCATION

Bob Jones University	<i>09/2018 - 05/2022</i>
B.S. in Mathematics	GPA: 4.0
B.S. in Computer Science	GPA: 4.0
 Clemson University	 <i>09/2022 - Present</i>
PhD in Mathematics under Hui Xue (in progress)	GPA: 4.0

RESEARCH INTERESTS

I work in analytic number theory. Recently, my work has involved very explicit computations of various objects relating to modular forms. This includes work related to trace formulas, the asymptotic behavior of the Hecke operators, and the asymptotic behavior of Fourier coefficients of modular forms. I also have some interests in harmonic analysis, dynamical systems/ergodic theory, and in theoretical computer science.

RESEARCH PAPERS

- **Equidistribution of CM points and RM curves** Erick Ross and Xui Xue. *Submitted 05/2026.*
- **Are sets with a given multiplicative structure guaranteed to have a density?** Erick Ross. *To appear in Acta Arithmetica.*
- **Transcendence of zeros of modular forms.** David Aiken, Erick Ross, Dmitriy Shvydkoy, Hui Xue. *Submitted 09/2025.*
- **Boundary CM points and class groups of small exponent.** David Aiken, Erick Ross, Dmitriy Shvydkoy, Hui Xue. *To appear in Proceedings of the AMS.*
- **Zeros of Even and Odd Period Polynomials** Grace Ko, Jennifer Mackenzie, Erick Ross, Hui Xue. *To appear in the Journal of Mathematical Analysis and Applications.*
- **Proportion of Atkin-Lehner sign patterns and Hecke Eigenvalue Equidistribution.** Erick Ross, Alexandre Van Lidth, Martha Rose Wolf, Hui Xue. *To appear in Mathematical Proceedings of the Cambridge Philosophical Society.*
- **Hecke Eigenvalue Equidistribution over the Newspace with Nebentypus.** Erick Ross. *To appear in the International Journal of Number Theory.*
- **Cramér α -random primes and the Fundamental Theorem of Arithmetic.** Erick Ross. *Submitted 05/2025.*

- **Dimension Sequences of Modular Forms.** Erick Ross. *Research in Number Theory* (2025), 14pp.
- **Non-repetition of second coefficients of Hecke polynomials.** Archer Clayton, Helen Dai, Tianyu Ni, Erick Ross, Hui Xue, Jake Zummo. *Indagationes Mathematicae* (2025), 22pp.
- **Asymptotics and sign patterns of Hecke polynomial coefficients.** Erick Ross, Hui Xue. *Canadian Mathematical Bulletin*, 68 (2025) 914-926.
- **On the average size of the eigenvalues of the Hecke operators.** William Cason, Akash Jim, Charlie Medlock, Erick Ross, Hui Xue. *Archiv der Mathematik*, 124 (2025) 255-263.
- **Nonvanishing of second coefficients of Hecke polynomials on the newspace.** William Cason, Akash Jim, Charlie Medlock, Erick Ross, Trevor Vilardi, Hui Xue. *International Journal of Number Theory*, 21 (2025) 1479-1512.
- **Signs of the second coefficients of Hecke polynomials.** Erick Ross, Hui Xue. *Illinois Journal of Mathematics*, 69 (2025) 323-351.
- **Newspaces with Nebentypus: An Explicit Dimension Formula and Classification of Trivial Newspaces.** Erick Ross. *Journal of Number Theory*, 278 (2026) 317-352.

RESEARCH TALKS

- 10/2026 (upcoming) - **Dimension Sequence of Modular Forms** *AMS (American Mathematical Society) Southeastern Sectional Meeting*, Kennesaw State University.
- 05/2026 - **Boundary CM points and class groups of small exponent.** *AFW (Automorphic Forms Workshop)*, Kansas State University.
- 05/2026 - **Equidistribution of Hecke eigenvalues.** *AFW (Automorphic Forms Workshop)*, Kansas State University.
- 04/2026 - **Boundary CM points and Class Groups of Small Exponent.** *TORA (Texas-Oklahoma Representations and Automorphic Forms) conference*, University of Oklahoma.
- 04/2026 (upcoming) - **Period polynomial zeros lie on the circle of symmetry.** *SERMON (SouthEastern Regional Meeting on Numbers) conference*, High Point University.
- 03/2026 - **Hecke Eigenvalue Equidistribution over the Newspaces with Nebentypus.** *AMS (American Mathematical Society) Southeastern Sectional Meeting*, Georgia Southern University.
- 03/2026 - **Boundary CM points and Class Groups of Small Exponent.** *Gainesville International Number Theory Conference*, University of Florida.
- 12/2025 - **Zeros of even and odd period polynomials lie on the circle of symmetry.** *PANTS (Palmetto Number Theory Series) conference*, Clemson University.

- 12/2025 - **Cramér α -random primes and the Fundamental Theorem of Arithmetic.** *GSS (Graduate Student Seminar)*, Clemson University.
- 11/2025 - **Dimension Sequences of Modular Forms.** *BARD (Bayou Arithmetic Research Day)*, Louisiana State University (Zoom).
- 11/2025 - **Cramér α -random primes and the Fundamental Theorem of Arithmetic.** *MSS (Mathematical Sciences Symposium)*, Bob Jones University.
- 10/2025 - **What is the asymptotic behavior of the Hecke polynomial coefficients?** *Maine-Québec Number Theory Conference*, University of Maine.
- 09/2025 - **Asymptotics and Sign Patterns for the Hecke Polynomial Coefficients.** *GSS (Graduate Student Seminar)*, Clemson University.
- 05/2025 - **When exactly will the newspaces exist?** *AFW (Automorphic Forms Workshop)*, University of North Texas.
- 03/2025 - **When precisely will the newspaces of modular forms exist?** *AMS (American Mathematical Society) Southeastern Sectional Meeting*, Clemson University.
- 03/2025 - **How large can we expect the newspaces to be?** *GSS (Graduate Student Seminar)*, Clemson University.
- 12/2024 - **How large are the eigenvalues of the Hecke operators?** *PANTS (Palmetto Number Theory Series) conference*, University of South Carolina.
- 09/2024 - **Nonvanishing of second coefficients of Hecke polynomials on the newspace.** *PANTS (Palmetto Number Theory Series) conference*, Wake Forest University.
- 09/2024 - **How large are the Hecke eigenvalues and corresponding Fourier coefficients?** *GSS (Graduate Student Seminar)*, Clemson University.
- 05/2024 - **Nonvanishing of the Second Coefficient of General Hecke Polynomials.** *AFW (Automorphic Forms Workshop)*, Oklahoma State University.
- 04/2024 - **Eigenbasis Decomposition of Hecke Eigenform Products via the Rankin-Selberg Method.** *SERMON (SouthEastern Regional Meeting on Numbers) conference*, Wofford College.
- 02/2024 - **From Tori to Elliptic Curves.** *DECAL (Data sECurity and mAchine Learning) Research Workshop*, Clemson University.

TEACHING EXPERIENCE

Instructor of Record at Clemson University

09/2023 - Present

- Spr. 2026 - MATH 2070: Business Calculus II.
- Spr. 2026 - STAT 2301: Statistical Methods I Lab.
- Fall 2025 - MATH 2070: Business Calculus II. *Principal Instructor of Record.*
- Spr. 2025 - MATH 2070: Business Calculus II.

- Fall 2024 - MATH 1060: Calculus of One Variable I.
- Spr. 2024 - MATH 1020: Business Calculus I.
- Fall 2023 - MATH 1040: Precalculus and Introductory Differential Calculus.

Teaching Assistant at Clemson University

09/2022 - 05/2023

- Spr. 2023 - MATH 3110: Linear Algebra.
- Fall 2022 - STAT 2301: Statistical Methods I Lab.

SERVICE

- 03/2026 - Organized a Special Session on Modular Forms at the 2026 AMS Sectional Meeting.
- 12/2025 - Volunteered at the PANTS XL conference.
- 09/2025 - Served as a speaker at the Clemson University Math Club.
- Summer 2025 and Summer 2024 - Supervised REU students at Clemson University.
- Summer 2025 - Volunteered as a mathematics tutor for high school students.
- 04/2025 - Volunteered at the Clemson Math-In (a service to help Clemson undergraduates struggling in math).
- 04/2025 - Volunteered at the Clemson Calculus Challenge (a high school mathematics outreach).
- 04/2025 - Served as a speaker at the Mathematical Sciences Symposium at Bob Jones University.
- 03/2025 - Volunteered at the 2025 AMS Southeastern Sectional Meeting.
- 03/2025 and 03/2024 - Curated the problemset for the Bob Jones University programming competition.
- 02/2025 and 09/2023 and 09/2021 - Served as a panelist for the Mathematical Sciences Symposium at Bob Jones University.
- 10/2024 - Served as a STEM judge for the VEX Robotics high school competition.
- 10/2023 - Volunteered at the PANTS XXXVI conference.
- Spring 2022 - Organized a competitive programming seminar at Bob Jones University.
- Fall 2021 - Organized a Putnam exam preparation seminar at Bob Jones University.

UNDERGRADUATE AWARDS

- Highest Academic Achievement Award at Bob Jones University. (Awarded to the Senior graduating with the most credits with a 4.0 GPA.)

- Mathematics Award and Computer Science Award at Bob Jones University.
- Winner of the 2021 and 2022 Scholastic Bowls at Bob Jones University.
- Winner of the 2021 and 2022 CCSCSE Programming Contests.
- Second in the U.S. at the 2020 IEEEExtreme Programming Contest.
- Advancement to the 2021 and 2024 ICPC North American Championships.
- Winner of the 2021 Mugs to Donuts x Jane Street Math Challenge.